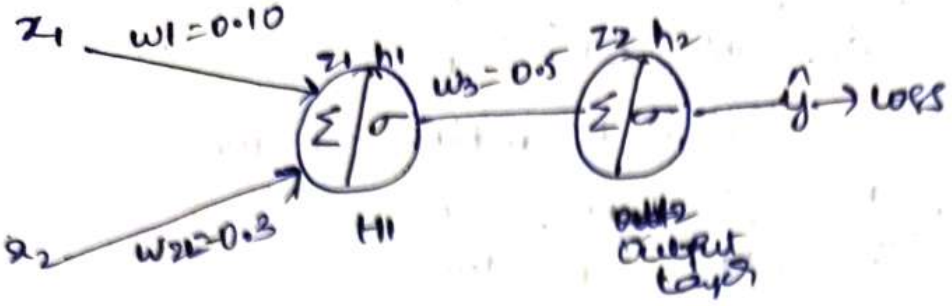

Dataset

x_1	x_2	y
1	2	1

Forward Propagation :

$$z_1 = w_1 x_1 + w_2 \cdot x_2$$
$$= (0.1)(1) + (-0.3) \cdot 2$$
$$z_1 = -0.5$$
$$h_1 = \sigma(z_1)$$
$$= \frac{1}{1+e^{-(-0.5)}}$$
$$h_1 = 0.3775$$

Layer 1

$$z_2 = h_1 \times w_3$$
$$= 0.3775 \times 0.5$$
$$z_2 = 0.1889$$
$$h_2 = \sigma(z_2)$$
$$= \frac{1}{1+e^{-0.189}}$$
$$h_2 = 0.547$$

$$\hat{y} = h_2$$
$$\hat{y} = 0.547$$

Loss MSE

$$= \frac{1}{n} \sum (y_i - \hat{y})^2$$
$$= (1 - 0.547)^2$$
$$\text{Loss} = 0.2052$$

Back Propagation:

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial h^1} \cdot \frac{\partial h^1}{\partial z^1} \cdot \frac{\partial z^1}{\partial w_1}$$

w_2 is treated as constant while cal w_1

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial h^1} \cdot \frac{\partial h^1}{\partial z^1} \cdot \frac{\partial z^1}{\partial w_2}$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial w_3}$$

$$\frac{\partial L}{\partial \hat{y}} \rightarrow L = \frac{\partial}{\partial \hat{y}} (y - \hat{y})^2$$

$$= 2(-1 + 0.547)^2$$

$$= 2(-0.453)$$

$$= -0.906$$

$$\frac{\partial \hat{y}}{\partial h^2} \rightarrow h^2 = \sigma(z^2)$$

$$\hat{y} = h^2$$

$$\frac{\partial \hat{y}}{\partial h^2} = \frac{\partial}{\partial h^2} \cdot h^2$$

$$= 1$$

$$\frac{\partial h^2}{\partial z^2} \Rightarrow h^2 = f(z^2)$$

$$h^2 = \sigma(z^2)$$

$$= \frac{\partial}{\partial z^2} \cdot \sigma(z^2)$$

$$= \sigma(z^2)(1 - \sigma(z^2))$$

$$= (0.547)(1 - 0.546)$$

$$\frac{\partial h^2}{\partial z^2} = 0.247$$

$$\frac{\partial z^2}{\partial h^1} = z^2 = h_1 \times w_2$$

$$= \frac{\partial}{\partial h_1} (h_1 \times w_2)$$

$$= w^2$$

$$\frac{\partial z^2}{\partial h^1} = -0.3$$

$$\frac{\partial h^1}{\partial z^1} = h^1 = \sigma(z^1)$$

$$= \frac{\partial}{\partial z^1} \cdot (\sigma(z^1))$$

$$= \sigma(z^1) \cdot (1 - \sigma(z^1))$$

$$\frac{\partial h^1}{\partial z^1} = 0.2349$$

$$\frac{\partial z^1}{\partial w_1} \quad z^1 = f(w_1)$$

$$= w_1 x_1 + w_2 x_2$$

$$\frac{\partial z^1}{\partial w_1} = 1$$

$$\frac{\partial z^1}{\partial w_2} = f(w_2)$$

$$= w_1 x_1 + w_2 x_2$$

$$\frac{\partial z^1}{\partial w_2} = 2.$$

$$\frac{dz^2}{dw_3} \qquad z^2 = h_1 \times w_3$$

$$= \frac{d}{d_3} \cdot (h' \times w_3)$$

$$\frac{dz^2}{dw_3} = 0.3755$$

$$\frac{dL}{dw_1} = -0.906 \times 1 \times 0.247 \times -0.3 \times 0.234 \times 1 = -0.124$$

$$\frac{dL}{dw_2} = -0.906 \times 1 \times 0.247 \times -0.3 \times 0.234 \times 2 = -0.251$$

$$\frac{dL}{dw_3} = -0.906 \times 1 \times 0.247 \times 0.3755 = -0.0816$$

Learning rate $\eta = 0.01$

$$w_{new} = w_{old} - \eta \times \frac{dL}{dw}$$

$$w_1 = 0.1 - (0.01) \times -0.124 = 0.1012$$

$$w_2 = -0.3 - (0.01) \times -0.254 = -0.297$$

$$w_3 = 0.5 - (0.01) \times -0.0816 = 0.500816$$